

Maxwell → Shielding

A step-by-step derivation: every classical calculator as one projection of a single kernel — and each step checked symbolically in Mathematica. The physics-architecture tower asserts a chain: Maxwell → $1/\rho$ → $\mathcal{D}_{ab}$ → a fork into magnetic and electrical doorways → the calculators → shielding. This notebook turns each arrow of that tower into a derivation. The aim is to stop trusting the classical calculators as formulas and to show them as named projections of one kernel, the near-field shadow of Maxwell. Mathematica is the engine and the check: it does the symbolic algebra, and it confirms each step before the prose claims it. Every code cell below ends in a verification that should read as **True** or as a matrix of zeros when the notebook is evaluated.

How to read this notebook

Each rung is one section, and each section follows the same five beats: the starting point, the move that gets made, the result, what the result means, and which calculators the result grounds. The prose states a step; the code cell beneath it confirms that step in the kernel; the display equation carries the result. A reader can follow the prose alone, run the cells to watch each claim verified, or read the closed plumbing cells for the figure code.

Conventions — the one object, stripped to its geometry

The derivation is carried in the **molecular frame**, with the source at the origin and the field point at displacement $\boldsymbol{\rho}$ of length ρ . The geometry is shown **unit-stripped**: the prefactors $\mu_0/4\pi$ (magnetic) and $1/4\pi\epsilon_0$ (electric) are dropped while the kernel is built, then restored at each doorway in Rung 5, where they carry the SI units. This keeps the geometry — the only thing shared across every calculator — uncluttered by constants that differ between the magnetic and electric branches.

The one object is the second derivative of the Coulomb potential, written $\mathcal{D}_{ab}$:

$$D_{ab} = \partial_a \partial_b \frac{1}{\rho} = \frac{3 \hat{\rho}_a \hat{\rho}_b - \delta_{ab}}{\rho^3}.$$

The setup cell defines the field point *coords*, the distance *rho*, the unit separation *rhat*, and builds *Dab* once by direct differentiation. Every later cell reuses these.

```
In[1]:= coords = {x, y, z};          (* field point; the source sits at the origin *)
rho = Sqrt[x^2 + y^2 + z^2];        (* separation distance *)
rhat = coords / rho;               (* unit separation vector *)
Dab = Table[D[1 / rho, coords[[a]], coords[[b]], {a, 3}, {b, 3}] // Simplify;
MatrixForm[Dab]
```

```
Out[5]//MatrixForm=
```

$$\begin{pmatrix} \frac{2x^2 - y^2 - z^2}{(x^2 + y^2 + z^2)^{5/2}} & \frac{3xy}{(x^2 + y^2 + z^2)^{5/2}} & \frac{3xz}{(x^2 + y^2 + z^2)^{5/2}} \\ \frac{3xy}{(x^2 + y^2 + z^2)^{5/2}} & \frac{-x^2 + 2y^2 - z^2}{(x^2 + y^2 + z^2)^{5/2}} & \frac{3yz}{(x^2 + y^2 + z^2)^{5/2}} \\ \frac{3xz}{(x^2 + y^2 + z^2)^{5/2}} & \frac{3yz}{(x^2 + y^2 + z^2)^{5/2}} & \frac{-x^2 - y^2 + 2z^2}{(x^2 + y^2 + z^2)^{5/2}} \end{pmatrix}$$

Rung 1 — Maxwell's source equations become Poisson equations for the potentials

The two source equations of electromagnetism are Gauss's law and Ampère's law. Gauss's law sets the divergence of the electric field by the charge density. Ampère's law (static) sets the curl of the magnetic field by the current density. $\nabla \cdot \mathbf{E} = \frac{\rho_{\text{ch}}}{\epsilon_0}$, $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$. Each field is rewritten through a potential. The electric field is minus the gradient of a scalar potential, which is exact because the static field is curl-free. The magnetic field is the curl of a vector potential, which is exact because the field is divergence-free. $\mathbf{E} = -\nabla \phi$, $\mathbf{B} = \nabla \times \mathbf{A}$. Substituting the scalar potential into Gauss's law turns it into a Poisson equation for ϕ . Substituting the vector potential into Ampère's law, and using the vector identity for the curl of a curl, turns it into a Poisson equation for each component of $\mathbf{A}$ once the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$ is imposed. $\nabla^2 \phi = -\frac{\rho_{\text{ch}}}{\epsilon_0}$, $\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$ ($\nabla \cdot \mathbf{A} = 0$). The cell verifies the two substitutions on generic fields. The electric check confirms that the divergence of $-\nabla \phi$ is $-\nabla^2 \phi$. The magnetic check confirms the curl-of-curl identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$, after which the gauge condition removes the first term.

```
In[6]:= (* electric: E = -grad phi, so Gauss div E = rho/eps0 becomes Poisson *)
phi = pot[x, y, z];
Efield = -Grad[phi, coords];
electricCheck = Simplify[Div[Efield, coords] + Laplacian[phi, coords]] == 0;

(* magnetic: B = curl A, and curl curl A = grad(div A) - Laplacian A *)
Avec = {ax[x, y, z], ay[x, y, z], az[x, y, z]};
magneticCheck = Simplify[Curl[Curl[Avec, coords], coords]
  - (Grad[Div[Avec, coords], coords] - Laplacian[Avec, coords])] == {0, 0, 0};

{"div(-grad phi) == -Laplacian phi" -> electricCheck,
 "curl curl A == grad(div A) - Laplacian A" -> magneticCheck}

Out[11]= {div(-grad phi) == -Laplacian phi -> True,
 curl curl A == grad(div A) - Laplacian A -> True}
```

The meaning is that both potentials obey the same operator, the Laplacian, against their own source. The electric and magnetic problems are now one mathematical problem solved twice. This is the step that makes a single Green's function serve both branches (Jackson, *Classical Electrodynamics*, §6.3). → grounds: nothing yet computes here; this rung sets up the shared operator that the rest of the chain solves.

Rung 2 — The Coulomb Green's function $1/\rho$ — the waist

The Poisson operator is inverted by its Green's function, the field of a unit point source. The Green's function satisfies the Laplacian against a point source of strength -4π , and its solution is $1/\rho$.

$\nabla^2 G = -4\pi \delta^3(\rho)$, $G(\rho) = \frac{1}{\rho}$. The cell confirms $1/\rho$ is a solution in two parts. Away from the origin

the Laplacian of $1/\rho$ is zero, so $1/\rho$ is harmonic everywhere the source is not. The source itself is recovered as a flux: the surface integral of $\nabla(1/\rho)$ over any sphere centred on the origin is -4π , independent of the radius, which is the integrated form of the delta source.

```
In[12]:= laplaceCheck = Simplify[Laplacian[1/rho, coords]]; (* 0 away from the origin *)
```

```
(* flux of grad(1/rho) through a sphere of radius R -- the delta source *)
gradG = Grad[1/rho, coords];
onSphere = {R Sin[th] Cos[ph], R Sin[th] Sin[ph], R Cos[th]};
flux = Integrate[
  (gradG . (coords/rho) /. Thread[coords -> onSphere]) R^2 Sin[th] // Simplify,
  {th, 0, Pi}, {ph, 0, 2 Pi}, Assumptions -> R > 0];
```

```
{"Laplacian[1/rho] off origin" -> laplaceCheck, "sphere flux of grad(1/rho)" -> flux}
```

```
Out[16]=
```

```
{Laplacian[1/rho] off origin -> 0, sphere flux of grad(1/rho) -> -4 Pi}
```

This is the waist of the whole chain. Both the scalar potential ϕ and the vector potential $\mathbf{A}$ are built by convolving their source against this one function $1/\rho$. The electric potential is $1/\rho$ smeared over charge; the magnetic vector potential is $1/\rho$ smeared over current. The place where the electric and the magnetic become one geometry is here, in the single function they share (Jackson §1.7).→
grounds: the Coulomb calculator is $1/\rho$ itself, undifferentiated; every other calculator is a derivative of this object.

Rung 3 — The second derivative: the kernel $\$D_{\{ab\}}$

A localized source is read at a distance through the derivatives of $1/\rho$. The first derivative is the field: the gradient of $1/\rho$ is the Coulomb field direction, falling as $1/\rho^2$. The second derivative is the field's spatial variation: the Hessian of $1/\rho$ is the tensor the thesis calls $\$D_{\{ab\}}$.

$\partial_a \frac{1}{\rho} = -\frac{\hat{\rho}_a}{\rho^2}$, $D_{ab} = \partial_a \partial_b \frac{1}{\rho} = \frac{3 \hat{\rho}_a \hat{\rho}_b - \delta_{ab}}{\rho^3}$. The cell verifies both derivatives against their closed

forms. The first-derivative check confirms the field is $-\hat{\rho}/\rho^2$. The second-derivative check subtracts the claimed closed form $(3\hat{\rho}\hat{\rho} - \delta)/\rho^3$ from the directly differentiated Hessian and returns the zero matrix.

```
In[17]:= field = Grad[1/rho, coords] // Simplify; (* first derivative: the field *)
claimed = Table[(3 rhat[[a] x rhat[[b]] - KroneckerDelta[a, b]) / rho^3, {a, 3}, {b, 3}];
identityCheck = Simplify[Dab - claimed]; (* the zero matrix *)
```

```
{"grad(1/rho) == -rhat/rho^2" -> (Simplify[field + rhat / rho^2] === {0, 0, 0}),
 "Dab - (3 rhat rhat - delta) / rho^3" -> MatrixForm[identityCheck]}
```

```
Out[20]=
```

```
{grad(1/rho) == -rhat/rho^2 -> True, Dab - (3 rhat rhat - delta) / rho^3 ->  $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ }
```

The meaning is that one tensor object now stands behind every through-space measure. The Hessian of the Coulomb potential is the same object as the dipolar coupling tensor and the electric field gradient. Whatever the source — an induced moment or a charge — its through-space signa-

ture at a nucleus is read through this single tensor. → grounds: D_{ab} is the object the magnetic and electric calculators each contract; the next rung reads its geometry, and Rung 5 sends it through the two doorways.

Rung 4 — The geometry of D_{ab}

The kernel is a real symmetric 3×3 tensor, so it is fixed by its trace, its eigenvalues, and its eigenvectors. The trace is zero, so the tensor has no isotropic part and is pure anisotropy — the part NMR through-space effects are made of. Its eigenvalues are $+2/\rho^3$ along the separation and $-1/\rho^3$ in the two directions across it. Contracting the tensor onto a fixed axis gives the angular law $(3\cos^2\theta -$

$1)/\rho^3$, which vanishes at the magic angle.

$$\text{Tr } D = 0, \quad D = D^T, \quad \text{eig}(D) = \left\{ \frac{\frac{+2}{\rho^3}, -1}{\rho^3} \right\}, \quad \hat{n} \cdot D \cdot \hat{n} = \frac{3\cos^2\theta - 1}{\rho^3}, \quad \theta_{\text{magic}} = \arccos \frac{1}{\sqrt{3}} = 54.74^\circ.$$

The cell reads each property off D_{ab} directly. It confirms the tensor equals its transpose and that its trace simplifies to zero. It returns the eigenvalues, which are the two $-1/\rho^3$ values and the single $+2/\rho^3$ value. It contracts D_{ab} onto the z-axis, substitutes the polar angle, and confirms the contraction equals $(3\cos^2\theta - 1)/\rho^3$, whose root is the magic angle.

```
In[21]:= symmetric = Dab === Transpose[Dab];
traceless = Simplify[Tr[Dab]] === 0;
{evals, evecs} = Eigensystem[Dab];

(* angular law: contract with a fixed axis along z; th = angle from that axis *)
nAxis = {0, 0, 1};
angular = Simplify[nAxis . Dab . nAxis /. {x -> r Sin[th], y -> 0, z -> r Cos[th]}];
magicCheck = Simplify[angular - (3 Cos[th]^2 - 1) / r^3, r > 0] === 0;
theta0 = ArcCos[1 / Sqrt[3]];

{"symmetric" -> symmetric, "traceless" -> traceless,
 "eigenvalues" -> Simplify[evals],
 "n.D.n == (3 cos^2 th - 1) / r^3" -> magicCheck,
 "magic angle (deg)" -> N[theta0 / Degree]}
```

```
Out[28]= {symmetric -> True, traceless -> True,
 eigenvalues -> {-1/(x^2 + y^2 + z^2)^(3/2), -1/(x^2 + y^2 + z^2)^(3/2), 2/(x^2 + y^2 + z^2)^(3/2)},
 n.D.n == (3 cos^2 th - 1) / r^3 -> True, magic angle (deg) -> 54.7356}
```

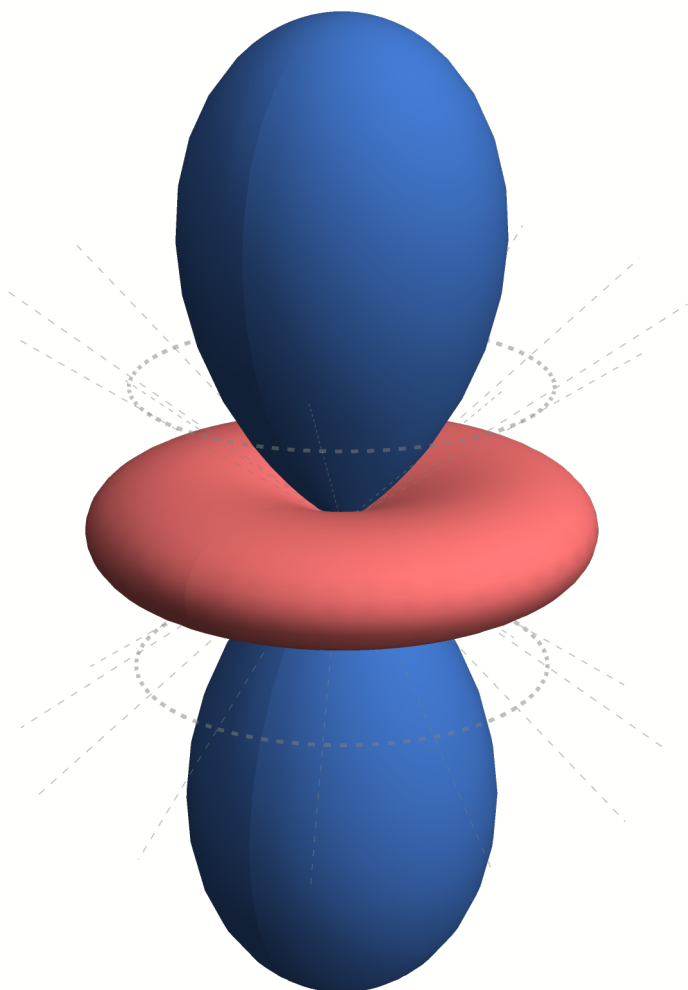
The figure draws the shape these numbers describe. The radial part falls as $1/\rho^3$, steep and near-field. The angular part is the double-lobed function $3\cos^2\theta - 1$: positive along the axis, where the eigenvalue is $+2/\rho^3$, and negative in the equatorial band, where the eigenvalue is $-1/\rho^3$. The two signs meet at the cone $\theta = 54.74^\circ$, where the kernel passes through zero. The plot shows the axial lobes in blue, the equatorial torus in red, and the magic-angle cone as the dashed ruling between

them.

```
In[29]:= magicAngle = ArcCos[1 / Sqrt[3]];
lobes = SphericalPlot3D[Abs[3 Cos[t]^2 - 1], {t, 0, Pi}, {pp, 0, 2 Pi},
  ColorFunction -> Function[{xx, yy, zz, t, pp},
    If[3 Cos[t]^2 - 1 >= 0, RGBColor[0.20, 0.36, 0.62], RGBColor[0.78, 0.36, 0.36]]],
  ColorFunctionScaling -> False, Mesh -> None, PlotPoints -> 45,
  Boxed -> False, Axes -> False, PlotRange -> All, ImageSize -> 380];
nodeRing[sgn_] := Line[Table[
  {Sin[magicAngle] Cos[pp], Sin[magicAngle] Sin[pp], sgn Cos[magicAngle]},
  {pp, 0, 2 Pi, 2 Pi / 60}]];
ruling[sgn_] := Table[Line[{{0, 0, 0},
  1.6 {Sin[magicAngle] Cos[pp], Sin[magicAngle] Sin[pp], sgn Cos[magicAngle]}},
  {pp, 0, 2 Pi - 0.01, 2 Pi / 12}]];
coneArt = {Gray, Opacity[0.5], Dashing[{0.01, 0.012}],
  ruling[1], ruling[-1], {Thick, nodeRing[1], nodeRing[-1]}};
Quiet@Show[lobes, Graphics3D[coneArt],
  PlotLabel -> Style["the shape of D_ab (node at 54.74 deg)", 12],
  ViewPoint -> {2.0, 1.4, 0.9}, PlotRange -> All]
```

Out[34]=

the shape of D_ab (node at 54.74 deg)



The meaning is that the entire angular content of every through-space NMR effect is this one function. Whatever wears it — the McConnell scalar, the dipolar coupling, the electric field gradient —

carries the same lobes and the same node. The thesis never collapses this rank-2 shape to a number; the next rungs keep it intact.→ grounds: the magic angle and the $(3\cos^2\theta - 1)/\rho^3$ law are the geometric content the angular calculators report; the radial $1/\rho^3$ is their distance dependence.

Rung 5 — The fork: two doorways, one kernel

The kernel reaches the nucleus by two routes. A magnetic source contracts D_{ab} with an induced moment; an electric source contracts the same D_{ab} with a charge. The two branches differ only in what is contracted and in the SI prefactor restored at the door. The tensor between them is identical.

Magnetic doorway — D_{ab} contracted with an induced moment

A point magnetic moment $\mathbf{m}$ sets up a vector potential proportional to $\mathbf{m} \times \hat{\rho} / \rho^2$. Its magnetic field is the curl of that vector potential. Away from the origin, that curl is exactly the kernel contracted with the moment, with the SI factor $\mu_0/4\pi$ restored.

$$A_{\text{dip}} = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\rho}}{\rho^2}, \quad B_a = (\nabla \times \mathbf{A})_a = \frac{\mu_0}{4\pi} D_{ab} m_b.$$

The cell builds the unit-stripped dipole vector potential from a constant moment $\mathbf{m}$, takes its curl to get the field, and subtracts the kernel contracted with the moment. The difference is the zero vector, so the dipole field is $D_{ab} m_b$.

```
In[35]:= m = {mx, my, mz}; (* induced point moment *)
Adip = Cross[m, coords] / rho^3;
(* dipole vector potential, mu0/4pi dropped *)
Bdip = Curl[Adip, coords] // Simplify; (* B = curl A, away from the origin *)
magneticFork = Simplify[Bdip - Dab . m] == {0, 0, 0};
{"B = curl(A_dip) equals D.m" -> magneticFork}

Out[39]:= {B = curl(A_dip) equals D.m -> True}
```

The induced moment is where the ring current enters: an aromatic ring in a field acquires a moment proportional to its susceptibility, and that moment's field at a nearby proton is read through D_{ab} . This is the doorway for **McConnell, Johnson-Bovey, Haigh-Mallion, the ring-susceptibility model, the Biot-Savart integration, and bond anisotropy** (Pople 1956; McConnell 1957).

Electric doorway — D_{ab} contracted with a charge

A point charge q sets up the Coulomb potential $1/\rho$ scaled by the charge. Its field is minus the gradient of that potential. The gradient of the field — the electric field gradient — is the same kernel D_{ab} , with the SI factor $1/4\pi\epsilon_0$ restored and the charge in place of the moment.

$$\varphi = \frac{1}{4\pi\epsilon_0} \frac{q}{\rho}, \quad \partial_a E_b = -\frac{q}{4\pi\epsilon_0} D_{ab}.$$

The cell builds the unit-stripped charge potential, takes minus its gradient for the field, then differentiates the field once more to form the field-gradient tensor. Subtracting $-q$ times the kernel gives the zero matrix, so the field gradient is the same D_{ab} the magnetic branch used.

```

In[40]:= phiq = q / rho; (* point-charge potential, 1/4pi eps0 dropped *)
Eq = -Grad[phiq, coords]; (* E = -grad phi *)
gradE = Table[D[Eq[[b]], coords[[a]], {a, 3}, {b, 3}] // Simplify; (* field gradient *)
electricFork = Simplify[gradE - (-q) Dab] === ConstantArray[0, {3, 3}];
{"grad(E) equals -q D" → electricFork}

Out[44]= {grad(E) equals -q D → True}

```

The charge and its field gradient are where the electrostatic calculators enter: a charged or polar site sets a field at the nucleus, and the gradient of that field is read through $\$D_{\{ab\}}$. This is the doorway for **Coulomb, APBS, the MOPAC-derived measures, EEQ, WaterField, and the π -quadrupole** (Slichter, *Principles of Magnetic Resonance*).

The fork is the central claim made concrete: the magnetic field of a moment and the field gradient of a charge are the same tensor, contracted with different sources. H-bond and dispersion measures ride the seam between the two doorways.→ *grounds: the two branches of the calculator library, each a contraction of the one kernel.*

Rung 6 — Back to shielding: the Ramsey sum

Shielding is the field the electron currents make when the applied field stirs them up (Ramsey 1950). The through-space part of the shielding tensor at nucleus i is the sum, over nearby sources s , of each source's kernel weighted by its intensity. The magnetic mechanisms add a field at the nucleus; the electric mechanisms perturb the electronic response. Both enter as a contraction of $\$D_{\{ab\}}$. $\sigma_{ab}(i) \approx \sum_s I_s D_{ab}(r_{is})$. The cell places two sources at different sites, each contributing its own kernel about the nucleus, and forms their weighted sum. The sum is still symmetric and still traceless, so the rank-2 anisotropic structure survives the summation: many sources add up to one tensor of the same shape, never to a scalar. The cell then contracts a single axial source onto its own axis and recovers the ring-current angular factor, the McConnell law $(1 - 3\cos^2\theta)/\rho^3$, which is minus the contraction of $\$D_{\{ab\}}$ (McConnell 1957).

```

In[45]:= Dsource[r0_] := With[{u = coords - r0, n = Sqrt[(coords - r0) . (coords - r0)]},
  Table[(3 (u[[a]] / n) (u[[b]] / n) - KroneckerDelta[a, b]) / n^3, {a, 3}, {b, 3}]];
sigma = Ione Dsource[{a0, 0, 0}] + Itwo Dsource[{0, b0, 0}]; (* sum_s I_s D_ab(r_is) *)
sumSymmetric = Simplify[sigma - Transpose[sigma]] === ConstantArray[0, {3, 3}];
sumTraceless = Simplify[Tr[sigma]] === 0;

(* McConnell ring-current factor: the axial diagonal of D *)
mcc = Simplify[({0, 0, 1} . Dsource[{0, 0, 0}] . {0, 0, 1})
  /. {x → r Sin[th], y → 0, z → r Cos[th]}];

{"sum stays symmetric" → sumSymmetric, "sum stays traceless" → sumTraceless,
  "McConnell factor n.D.n" → mcc}

```

```

Out[50]= {sum stays symmetric → True, sum stays traceless → True,
  McConnell factor n.D.n →  $\frac{1 + 3 \cos[2 \text{ th}]}{2 (r^2)^{3/2}}$ }

```

The loop closes here. Maxwell's source equations became Poisson equations; the Poisson operator was inverted by $1/\rho$; the second derivative of $1/\rho$ was the kernel $\$D_{\{ab\}}$; the kernel's geometry

gave the magic angle and the $(3\cos^2\theta - 1)/\rho^3$ law; the kernel forked into a magnetic and an electric doorway; and the doorways sum back into the shielding tensor. Each calculator computes one term of the final sum. → grounds: the whole library at once — the sum is the calculators added together, each term one source read through the one kernel.

The chain, closed

The tower is now a theorem rather than a slogan. Maxwell → $1/\rho$ → $\$D_{\{ab\}}\$$ → the fork → the calculators → shielding holds at every rung, and each rung is checked above in the kernel. The classical calculators are not formulas the thesis trusts; they are the only thing they could have been, named projections of one geometry that is the near-field shadow of Maxwell.

References

Jackson, J. D. (1999). *Classical Electrodynamics*, 3rd ed. Wiley. §1.7 (Green's function of the Laplacian), §5.6 (magnetic dipole field), §6.3 (potentials and gauge). **McConnell, H. M.** (1957). Theory of nuclear magnetic shielding in molecules. *Journal of Chemical Physics* **27**, 226–229. The $(1 - 3\cos^2\theta)/r^3$ ring-current shift from a point anisotropic susceptibility. **Pople, J. A.** (1956). Proton magnetic resonance of hydrocarbons. *Journal of Chemical Physics* **24**, 1111. The point-dipole model of the aromatic ring current. **Ramsey, N. F.** (1950). Magnetic shielding of nuclei in molecules. *Physical Review* **78**, 699–703. The shielding tensor as the current response to the applied field. **Slichter, C. P.** (1990). *Principles of Magnetic Resonance*, 3rd ed. Springer. The electric field gradient and the quadrupolar interaction.